

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Both needed for one mark X = (gill) filaments Y = (gill) arch/bar	1			1		1
		(ii)		{Filter/sieve/remove/catch} {food/particles/solids NOT molecules} /protection from debris/preventing damage to the gill filaments (1)		1		1		
		(iii)		A. normally water separates {filaments/lamellae/plates/gills}/Gills dry out (when out of water) (1) B. {Filaments/lamellae/plates/gills} {stick together/collapse} (1) C. Reduced Surface Area for {diffusion/gas exchange} (1) D. Water needed for <u>diffusion</u> (of gases)/{oxygen/gases} need to be in solution to <u>diffuse</u> (1)	4			4		
	(b)			Accept reverse argument for any point A. Countercurrent more efficient (1) B. Maintains {diffusion/concentration} gradient/or description of/equilibrium not met (1) C. Across {entire/whole} {gill/filament/lamellae/plate} (1) D. {Higher/more} {saturation/oxygen absorbed} (1) Accept correct references to higher percentages in counter current/use of values e.g. 80% compared to 50%	4			4		
	(c)	(i)		C = parallel flow and A = countercurrent flow (1)		1		1		
		(ii)		Correct insertion of arrows on A <u>only</u> (1) NOT ECF	1			1		